

Volume II Quick Start

Deep Dive into DeepSeek — Volume II Quick Start

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Foundation Books Project

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A handoff may change the request's *representation* or *owner*.

It may **not** change its *identity* or *meaning*.

- Release pages and model cards state claims about behavior, interfaces, architecture, and performance. This quick start calls each one a **stated capability**.
- Volume I explained the model and training mechanisms behind those capabilities. Volume II asks what must happen for one of them to stay *valid* and *correctly scoped* in a working stack.
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Goal: install REQUEST-TRACE — one logical request, three separately sourced evidence lanes — plus the carried objects and the identity relation every later systems chapter is audited against.

1. A capability becoming a system

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2. Objects that must stay attached

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4. From concept to code

Roadmap

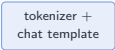
1. A capability becoming a system
2. Objects that must stay attached
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4. From concept to code
5. Carrying it forward

A capability becoming a system

From a stated capability to an owned object

Definition (Stated capability)

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tokenizer +
chat template

a tokenizer and chat template produce model input.

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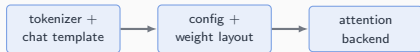
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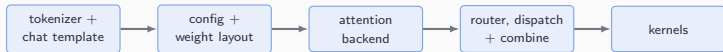
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Every arrow in that chain is a handoff, and every handoff is a place a capability can quietly stop being the thing that was claimed.

One logical request, one fixed prompt

The running example is REQUEST-TRACE. The observed run used this prompt, reproduced exactly:

Return a JSON object with keys status and note. Set status to ok and keep note under six words.

- The wording specifies a response *shape* and stipulates the value `ok`; the short `note` is otherwise unconstrained.

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The fixture is deliberately boring. A running example earns its keep by being reproducible and checkable, not by being impressive.

Three lanes, one label

Lane	Fixed record	What the record supports
/OBSERVED	21 input ids; output ids [185, 185]; TP2/EP2; dynamic FP8 weights; BF16 KV cache; DeepEP normal mode; FlashMLA; page size 64	A two-H200 capability-and-handoff record for the listed state

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Three independently sourced inputs sharing one request label. Sharing a label does not make them one execution record.

The rule that keeps the lanes honest

Invariant (Lane discipline)

A field may move to another lane only as a labeled worked projection. It may never silently replace the canonical request.

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The Systems Foundations chapter turns this plain-language rule into formal trace and handoff contracts.

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This is why the observed lane records “dynamic FP8 weights, FP8 dispatch payloads with FP32 scales, BF16 KV cache” instead of a single precision adjective. [OBSERVED: job 371933 protocol record]

Objects that must stay attached

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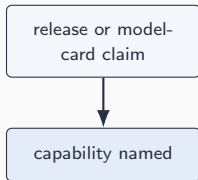
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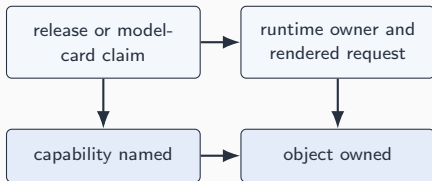
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Each row names state that later chapters may *transform* but must not silently *replace*.

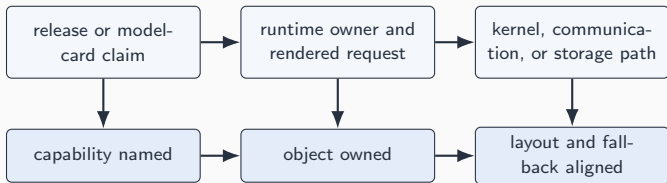
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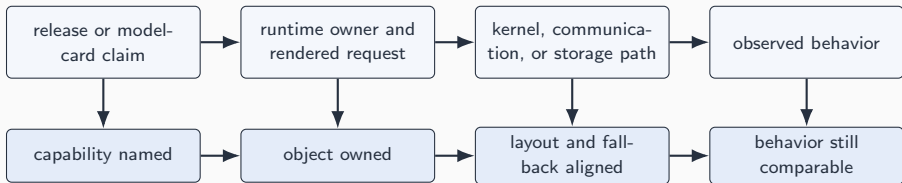
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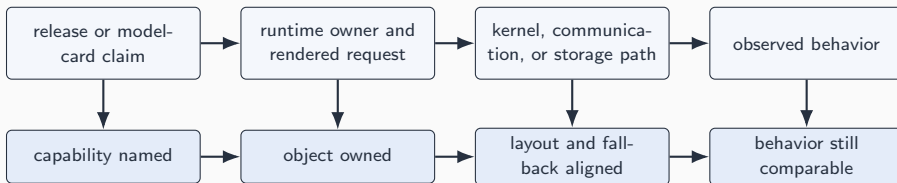
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Top row: the object moving into infrastructure. Bottom row: the check that keeps the handoff attached to the same capability.

This is a handoff diagram, not a new architecture. Each check identifies the object, its owner, and the state the next component may safely rely on.

The trace across Volume II

Active branches, with their boundaries

Active branch	State carried forward	Handoff boundary
Observed attention	108 FlashMLA calls; BF16 cache shape [47746, 64, 1, 576], stride [36864, 576, 576, 1], page size 64	scheduler admission and block-table values stay outside

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Inactive mechanisms add no state to the observed lane.

Two jobs, two denominators

Record	What it bounds	Stamp
job 371933	request-scoped identity and ordered handoffs for one request	[OBSERVED: capability and handoff trace]

- The throughput record stays a *separate* observed anchor. Each job keeps its own denominator.

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Two observed records about the same stack are still two records. Merging their denominators is the most tempting mistake in this volume.

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Later chapters apply this audit with named objects, evidence, interpretations, owners, release conditions, oracles, and measurement boundaries.

From concept to code

From concept to code: the ARCHITECTURE lane's coordinates

The ARCHITECTURE lane is not a paraphrase — it is a pinned configuration file.

```
from DeepSeek-V3/inference/configs/config_671B.json
```

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"dim": 7168, "n_layers": 61, "n_dense_layers": 3, "n_heads": 128,  
"n_routed_experts": 256, "n_shared_experts": 1, "n_activated_experts": 8,  
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These fields give the lane its shapes — and nothing else. They do not say that the logical request executed on V3, selected a backend, produced output, or achieved a latency. [INSPECTED:

DeepSeek-V3 inference/configs/config_671B.json]

From concept to code: combine returns the row

“Combine returns token rows to the request” has a literal home in the reference MoE forward pass.

```
from DeepSeek-V3/inference/model.py

weights, indices = self.gate(x)
y = torch.zeros_like(x)
for i in range(self.experts_start_idx, self.experts_end_idx):
    idx, top = torch.where(indices == i)
    y[idx] += expert(x[idx]) * weights[idx, top, None]
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`idx` is the origin row identity and `top` is the route slot. The expert output re-enters the request at its *own* row, scaled by its *own* gate weight. Any faster transport must reproduce exactly that. [INSPECTED: DeepSeek-V3 inference/model.py]

Carrying it forward

Concept-to-code map for the quick start

- lane definitions, provenance classes, claim boundaries →

`book/_metadata/volume_ii_request_trace.json`

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Change the representation, change the owner — keep the identity.

Next: Systems Foundations develops the CUDA, fabric, distributed-systems, pipeline, and measurement vocabulary used throughout Volume II, and turns this quick start's plain-language rule into the volume's formal trace and handoff contracts.